CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 7: Indices (Exponents)

Topic-wise short notes with solved examples

The laws of indices for positive, zero, negative and fractional exponents, simplifying expressions with indices and solving exponential equations.

1 Meaning of an index

- $a^n = a \times a \times \ldots \times a$ (n factors). Here a is the **base** and n is the **index** (exponent or power).
- **Zero index**: $a^0 = 1$ for $a \neq 0$.
- **Negative index**: $a^{-n} = \dfrac{1}{a^n}$, so $\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$.
- **Fractional index**: $a^{\frac{1}{n}} = \sqrt[n]{a}$ and $a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$.

Example 1.1 Evaluate $8^{\frac{2}{3}}$, $\left(\dfrac{27}{8}\right)^{-\frac{2}{3}}$ and $(0.01)^{-\frac{1}{2}}$.

SOLUTION

$8^{\frac{2}{3}} = (2^3)^{\frac{2}{3}} = 2^2$

$\left(\dfrac{27}{8}\right)^{-\frac{2}{3}} = \left(\dfrac{8}{27}\right)^{\frac{2}{3}} = \left(\dfrac{2}{3}\right)^2$

$(0.01)^{-\frac{1}{2}} = (10^{-2})^{-\frac{1}{2}} = 10$

4, $\dfrac{4}{9}$ and 10

2 Laws of indices

- These hold for all rational indices (with non-zero bases where a division occurs).

Law	Statement
Product	$a^m \times a^n = a^{m+n}$
Quotient	$a^m \div a^n = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Power of a product	$(ab)^n = a^n b^n$
Power of a quotient	$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$

- If $a^m = a^n$ and a is not 0, 1 or -1, then $m = n$.
- If $a^n = b^n$ and $n \neq 0$, then $a = b$ (for positive a, b).

Example 2.1 Simplify $\dfrac{2^5 \times 2^{-3} \times 4^2}{8}$.

SOLUTION

$\dfrac{2^5 \times 2^{-3} \times 2^4}{2^3} = 2^{5-3+4-3}$

$2^3 = 8$

Example 2.2 Simplify $\left(\dfrac{x^a}{x^b}\right)^{a+b} \times \left(\dfrac{x^b}{x^c}\right)^{b+c} \times \left(\dfrac{x^c}{x^a}\right)^{c+a}$.

SOLUTION

$x^{(a-b)(a+b)} \times x^{(b-c)(b+c)} \times x^{(c-a)(c+a)} = x^{a^2-b^2+b^2-c^2+c^2-a^2}$

$x^0 = 1$

Example 2.3 Simplify $(27)^{\frac{2}{3}} \times (16)^{-\frac{3}{4}} \div (125)^{-\frac{1}{3}}$.

SOLUTION

$9 \times \dfrac{1}{8} \div \dfrac{1}{5} = 9 \times \dfrac{1}{8} \times 5$

$\dfrac{45}{8}$

3 Working with roots

- Write every root as a fractional index, then use the laws.
- $\sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{ab}$ and $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$.

Example 3.1 Simplify $\sqrt[3]{x^2} \times \sqrt{x^3} \div x^{\frac{1}{6}}$.

SOLUTION

$x^{\frac{2}{3}} \times x^{\frac{3}{2}} \div x^{\frac{1}{6}} = x^{\frac{4+9-1}{6}}$

x^2

Example 3.2 Simplify $\dfrac{5^{n+3} - 6 \times 5^{n+1}}{9 \times 5^n - 5^n \times 2^2}$.

SOLUTION

Numerator: $5^n(125 - 30) = 95 \times 5^n$. Denominator: $5^n(9 - 4) = 5 \times 5^n$.

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4 Exponential equations

- An **exponential equation** has the unknown in the index.
- **Method**: write both sides as powers of the **same base**, then equate the indices.

- If the equation has terms like 2^{x+1} and 2^{x-1}, take 2^x common, or put $2^x = y$.

Example 4.1 Solve $2^{x+3} = 32$.

SOLUTION

$2^{x+3} = 2^5$, so $x + 3 = 5$

$x = 2$

Example 4.2 Solve $9^x = 27^{x-1}$.

SOLUTION

$3^{2x} = 3^{3x-3}$, so $2x = 3x - 3$

$x = 3$

Example 4.3 Solve $\left(\frac{2}{3}\right)^x = \left(\frac{27}{8}\right)^2$.

SOLUTION

$\left(\frac{27}{8}\right)^2 = \left(\frac{3}{2}\right)^6 = \left(\frac{2}{3}\right)^{-6}$

$x = -6$

Example 4.4 Solve $2^{x+2} + 2^x = 40$.

SOLUTION

$2^x(4 + 1) = 40$, so $2^x = 8$

$x = 3$

Example 4.5 Solve $3^{2x} - 10 \times 3^x + 9 = 0$.

SOLUTION

Let $y = 3^x$: $y^2 - 10y + 9 = 0$ gives $y = 1$ or $y = 9$.

$x = 0$ or $x = 2$

Example 4.6 If $2^x = 3^y = 6^{-z}$, show that $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$.

SOLUTION

Let each equal k. Then $2 = k^{\frac{1}{x}}$, $3 = k^{\frac{1}{y}}$ and $6 = k^{-\frac{1}{z}}$.

$2 \times 3 = 6$ gives $k^{\frac{1}{x}+\frac{1}{y}} = k^{-\frac{1}{z}}$.

$\frac{1}{x} + \frac{1}{y} = -\frac{1}{z}$, which is the required result

5 Common mistakes

- Writing $a^m \times a^n = a^{mn}$ or $(a^m)^n = a^{m+n}$.
- Taking a^{-n} to be negative. $2^{-3} = \frac{1}{8}$.
- Writing $a^0 = 0$.
- Applying the laws across different bases: $2^3 \times 3^2$ is not 6^5.
- Writing $(a+b)^n = a^n + b^n$.
- Confusing $a^{\frac{m}{n}}$ with $a^{\frac{n}{m}}$: the denominator of the index is the root.

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